

9/3/26

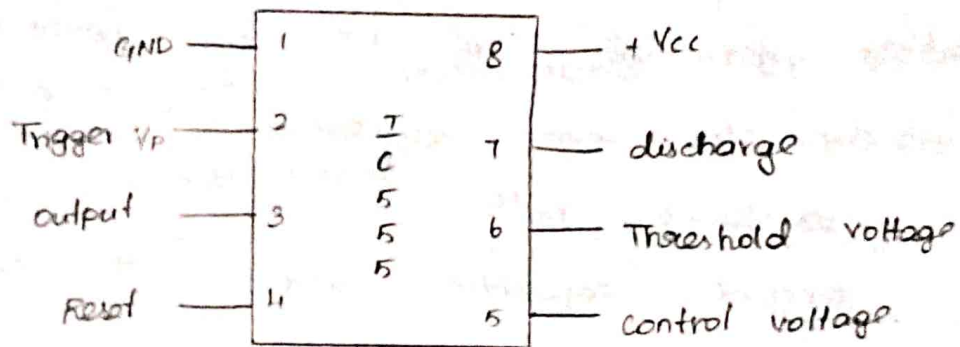
UNIT-IV.

TIMER AND PLL :

- 1) IC 555 timer.
- 2) Voltage controlled Oscillator (VCO)
- 3) Phase locked loop (PLL)

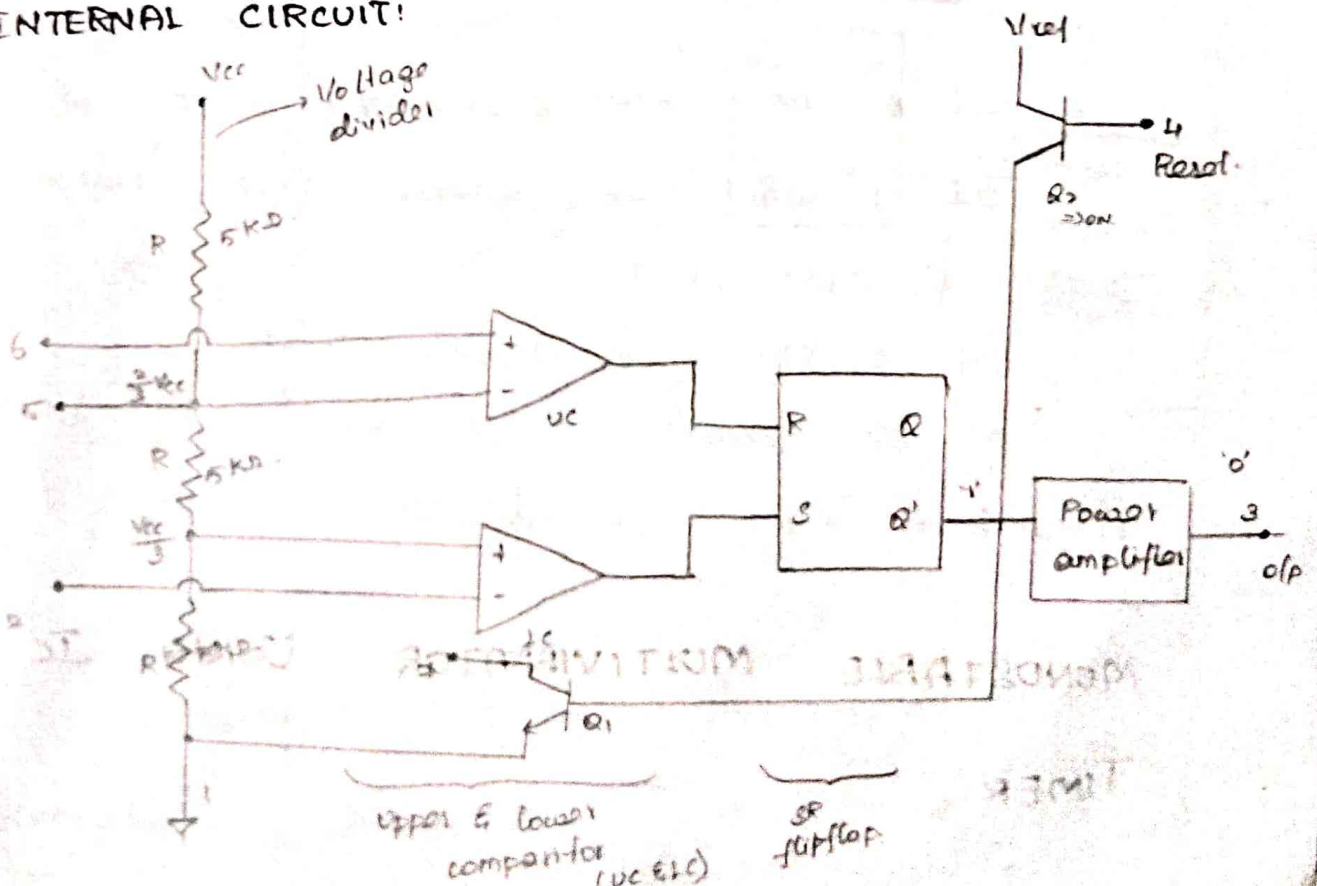
IC 555 Timer:

PIN DIAGRAM.



Supply voltage is 4.8 to 18 V.

INTERNAL CIRCUIT:



All resistors used in this IC is 5kΩ
that's why is called 555 timer.



$$V = \frac{V_{cc}}{R+R} (2R)$$

$$= \frac{V_{cc}}{3R} (2R)$$

$$V_1 = \frac{2}{3} V_{cc}$$



$$V_1 = \frac{V_{cc}}{3R} (2R)$$

$$= \frac{V_{cc}}{3}$$

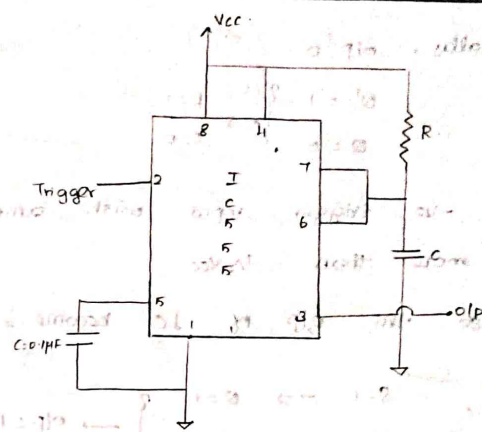
⇒ If you want to alter the voltage, and inserting terminal of IC we connect pin 5.
⇒ Capacitor can discharge via Q_1 , to create low resistance path.
connect capacitor across the collector terminal of transistor.



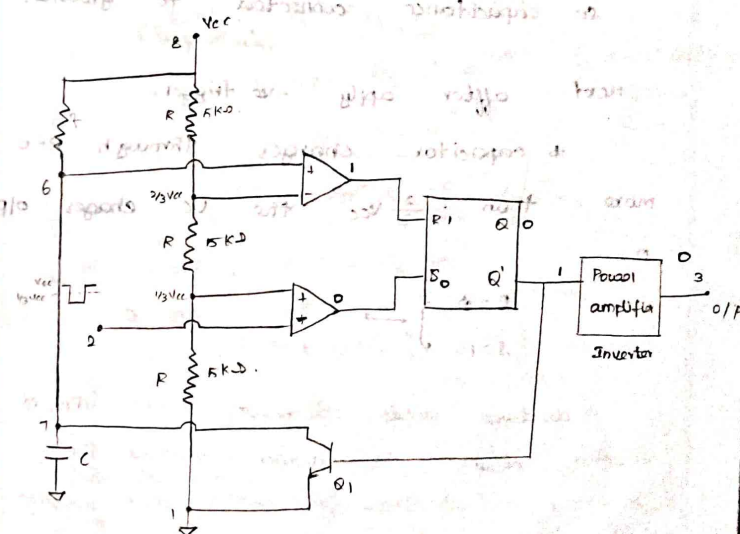
⇒ Reset - negative edge triggered pulse.
⇒ If Q_2 - ON make $Q_1 = 1$ o/p is 0. allow reset.
⇒ If we want to disable Reset signal.
Simply Q_2 make OFF
If Q_2 - OFF not allowing Reset
If we connect V_{cc} to Reset, Q_2 become
of Reset signal is disabled.

MONOSTABLE MULTIVIBRATOR USING IC 555

TIMER:



This performs monostable operation (or) one-shot multivibrator.



Modified circuit to perform monostable operation.

Initially, $o/p = 0$.

$$\left. \begin{matrix} Q' = 1 \\ Q = 0 \end{matrix} \right\} \rightarrow \begin{matrix} R = 1 \\ S = 0 \end{matrix}$$

Apply -ve trigger input with amplitude more than $\frac{1}{3} V_{cc}$.

so the o/p of IC become = 1.

$$\left. \begin{matrix} \text{after short duration} \\ \text{it returns 0.} \end{matrix} \right\} \rightarrow \begin{matrix} S = 1 \\ R = 1 \end{matrix} \rightarrow \left. \begin{matrix} Q = 1 \\ Q' = 0 \end{matrix} \right\} \rightarrow o/p = 1$$

initially, $Q_1 \rightarrow ON$ creates low resistance path so capacitance connected to ground. (discharged)

next after apply -ve trigger.

capacitor charges through $R=0$

more than $\frac{2}{3} V_{cc}$ the VC charges o/p 0.

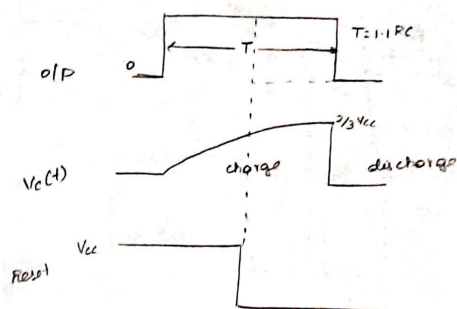
$$\left. \begin{matrix} R = 0 \\ S = 1 \end{matrix} \right\} \rightarrow \left. \begin{matrix} Q = 0 \\ Q' = 0 \end{matrix} \right\} \rightarrow o/p = 0$$

$\Rightarrow$ discharge sudden decrease bcz Q_1 of C directly connect to ground without P.

both $R=S=1$, so we need to apply short duration pulse with amplitude more than $\frac{1}{3} V_{cc}$, so $S=1$ will not to long time if changes to $S=0$.

$$\left. \begin{matrix} R = 1 \\ S = 0 \end{matrix} \right\} \rightarrow \left. \begin{matrix} Q = 1 \\ Q' = 1 \end{matrix} \right\} \rightarrow o/p = 0$$

if -ve trigger pulse is applied with amplitude more than $\frac{1}{3} V_{cc}$, the output of the IC will be 1.



Voltage across capacitance is expressed as.

$$\begin{aligned} V_c(t) &= V_f - (V_f - V_i) e^{-t/RC} \\ &= V_{cc} - (V_{cc} - 0) e^{-t/RC} \quad [\because \text{during charging } V_f = V_{cc} \\ &= V_{cc} - V_{cc} e^{-t/RC} \quad V_i = 0] \end{aligned}$$

$$V_c(t) = V_{cc} (1 - e^{-t/RC})$$

$$\text{at } t = T, V_c(t) = \frac{2}{3} V_{cc}$$

$$\frac{2}{3} V_{cc} = V_{cc} [1 - e^{-T/RC}]$$

$$e^{-T/RC} = 1 - \frac{2}{3} = \frac{1}{3}$$

take ln on b/s,

$$-\frac{T}{RC} = \ln \left(\frac{1}{3} \right)$$

$$+\frac{T}{RC} = -1.098$$

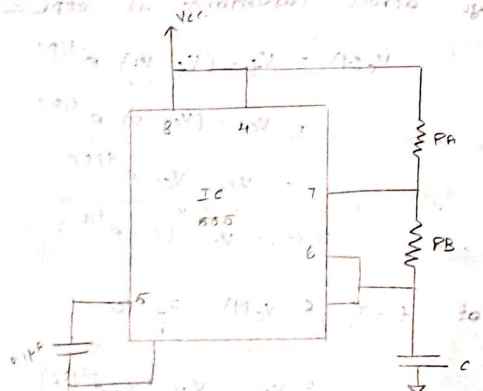
$$T = 1.1 RC$$

This is the time period of the o/p generated by the monostable multivibrator.

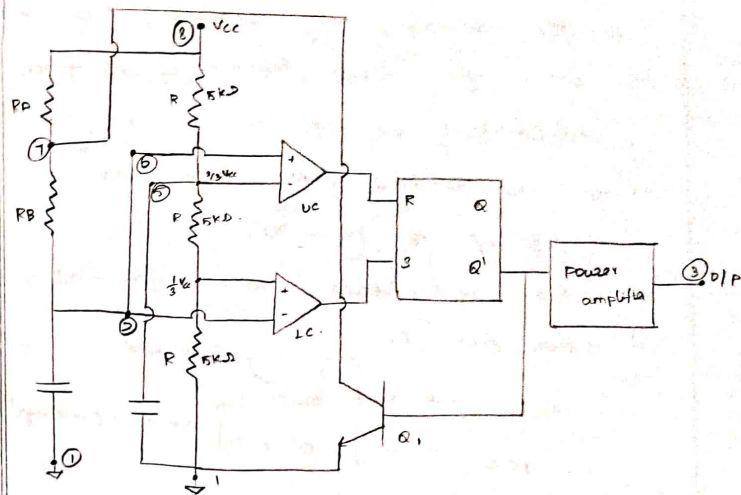
excitation table:

	Reset	Set	Reset	Set
	R	S	$\bar{Q}$	Q_{n+1}
0	0	0	0	0
0	0	1	0	1
1	0	0	1	0
0	0	0	1	1
1	1	1	x	x - invalid

Monostable operation using IC 555 timer.



Capacitor is charged through two resistors R_1 & R_2 .



If we don't want the pin 5 (control voltage) so we connect with ground through capacitor.

First: $UC \rightarrow \frac{0}{8} V_{cc}$ in +ve terminal. No voltage in -ve " so o/p of UC $\rightarrow 0$.

$LC \rightarrow \frac{1}{3} V_{cc}$ in +ve. No volt in -ve so o/p of LC $\rightarrow 1$.

$UC \rightarrow 0$ $R = 0$ } $Q = 1$ } $\rightarrow$ o/p '1'
 $LC \rightarrow 1$ $S = 1$ } $\bar{Q} = 0$ }
 (there no connection b/w C & ground).

If $\bar{Q} = 0 \rightarrow$ transistor (Q) is OFF, so

capacitor is connected to V_{cc} through R_A & R_B .

$\Rightarrow$ allows capacitor to get charge through

R_A & R_B .

so the voltage at capacitor $\uparrow$ so upto V_{cc} but, after charge beyond $\frac{2}{3}$ (or) $\frac{1}{3} V_{cc}$,
 * whenever it is crossing $\frac{1}{3} V_{cc}$,
 $LC \Rightarrow$ -ve terminal more than +ve.

so o/p of $LC \Rightarrow 0$.

But VC unchanged becoz $\frac{1}{3} V_{cc} \rightarrow \frac{2}{3} V_{cc}$
 it takes some time.

Now, $R=0 \left\{ \begin{array}{l} Q=1 \\ Q'=0 \end{array} \right\} \rightarrow o/p=1$ (unchanged)
 $LC \Rightarrow s=0 \left\{ \begin{array}{l} Q=1 \\ Q'=0 \end{array} \right\} \rightarrow o/p=1$ (unchanged)

* when crossing $\frac{2}{3} V_{cc}$,

VC $\Rightarrow$ +ve terminal more than -ve.

so o/p of VC $\Rightarrow 1$

$R=1 \left\{ \begin{array}{l} Q=0 \\ Q=1 \end{array} \right\} \rightarrow o/p=0$
 $s=0 \left\{ \begin{array}{l} Q=0 \\ Q=1 \end{array} \right\} \rightarrow o/p=0$

$\bar{Q}=1 \rightarrow$ transistor is ON so capacitor start discharging through RB.

* Now, whenever $\downarrow$ below $\frac{2}{3} V_{cc}$.

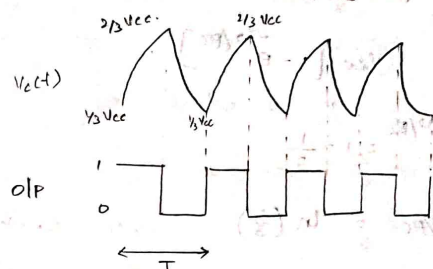
VC $\Rightarrow$ become +ve < -ve so o/p = 0.

whenever $\downarrow$ below $\frac{1}{3} V_{cc}$.

LC $\Rightarrow$ become -ve < +ve terminal so o/p = 1.

$R=0 \left\{ \begin{array}{l} Q=1 \\ Q=0 \end{array} \right\} \rightarrow o/p=1$

$\Rightarrow \bar{Q}=0$, (Q) transistor OFF, charging process occurs



General equation.

$$V_c(t) = V_f + (V_i - V_f) e^{-t/RC}$$

$$[-: T = t_{high} + t_{low}]$$

$t_1 \rightarrow$ capacitor : $0V \rightarrow \frac{2}{3} V_{cc}$.

$t_2 \rightarrow$ capacitor : $0V \rightarrow \frac{1}{3} V_{cc}$.

$$t_{high} = t_1 - t_2 = \frac{1}{3} \rightarrow \frac{2}{3} V_{cc} \text{ (charging time)}$$

$$V_c(t) = V_{cc} - (V_{cc} - 0) e^{-t/RC}$$

$$V_c(t) = V_{cc} [1 - e^{-t/RC}]$$

at $t = T_1$ $0 \rightarrow \frac{2}{3} V_{cc}$ t_1 $\frac{2}{3} V_{cc} \rightarrow 0$

$$\frac{2}{3} V_{cc} = V_{cc} [1 - e^{-T_1/RC}]$$

$$e^{-T_1/RC} = 1 - \frac{2}{3}$$

$$e^{-T_1/RC} = \frac{1}{3}$$

$$-T_1/RC = -1.098.$$

$$+T_1 = 1.1 RC.$$

$$\text{at } t \rightarrow T_2, \quad 0 \rightarrow \frac{1}{3} V_{CC}.$$

$$\frac{1}{3} V_{CC} = V_{CC} [1 - e^{-T_2/RC}]$$

$$e^{-T_2/RC} = 1 - \frac{1}{3}.$$

take ln,

$$-T_2/RC = \ln\left(\frac{2}{3}\right)$$

$$\frac{-T_2}{RC} = -0.405$$

$$T_2 = 0.41 RC.$$

$$t_{\text{high}} = T_1 - T_2.$$

$$t_{\text{high}} = t_{\text{on}}.$$

$$= (1.1 - 0.41) RC.$$

$$t_{\text{high}} = 0.69 RC. \Rightarrow T_{\text{on}} = 0.69 (R_A + R_B) C.$$

$$\therefore t_{\text{low}} \rightarrow \text{discharging time } t_2 - t_1. \quad \left(\frac{2}{3} V_{CC} \rightarrow \frac{1}{3} V_{CC}\right)$$

$$V_C(t) = V_f - (V_f - V_i) e^{-t/RC}.$$

$$[V_{CC} \rightarrow 0V]$$

$$= 0 - (0 - V_{CC}) e^{-t/RC}.$$

$$V_C(t) = V_{CC} e^{-t/RC}$$

$$\text{at } t \rightarrow t_1, \quad \frac{2}{3} V_{CC} \rightarrow 0$$

$$\therefore t_{\text{low}} \rightarrow \text{discharging time } (T_d).$$

$$V_C(t) = V_f - (V_f - V_i) e^{-t/RC} \quad \left(\frac{2}{3} V_{CC} \rightarrow \frac{1}{3} V_{CC}\right)$$

$$= 0 - \left(0 - \frac{2}{3} V_{CC}\right) e^{-t/RC}.$$

$$V_C(t) = \frac{2}{3} V_{CC} \cdot e^{-t/RC}.$$

$$\text{at } t = t_d.$$

$$\frac{1}{3} V_{CC} = \frac{2}{3} V_{CC} e^{-t_d/RC}.$$

$$\frac{1}{2} = e^{-t_d/RC}.$$

$$\ln(0.5) = -t_d/RC.$$

$$-0.693 = -\frac{t_d}{RC}.$$

$$\therefore T_d = 0.69 RC.$$

$$t_{\text{low}} = t_d = 0.69 R_B C.$$

$$T = T_{\text{high}} + T_{\text{low}}.$$

$$= T_c + T_d.$$

$$T = 0.69 [R_A + 2R_B] C.$$

$$f = \frac{1}{T} \approx \frac{1.44}{(R_A + 2R_B) C}.$$

To find Duty cycle: (b)

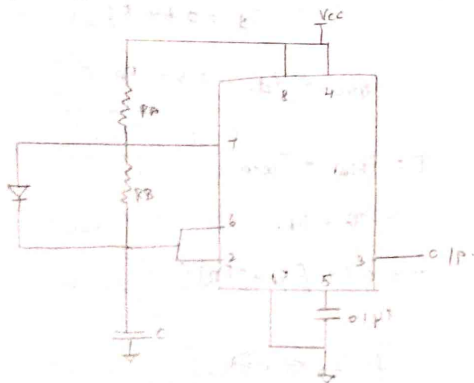
It is the ratio b/w ON time and total time period.

$$D = \frac{T_{ON}}{T} = \frac{0.69 (R_A + R_B) C}{0.69 (R_A + 2R_B) C}$$

$$D = \frac{R_A + R_B}{R_A + 2R_B}$$

This is suitable for duty cycle is greater than 50%.

To generate $< 50\%$, $= 50\%$ we modify the circuit.



The capacitor charge through R_A &

discharge through R_B .

$$T_{ON} = 0.69 (R_A) C \quad (\text{charging})$$

design an astable multivibrator using IC 555 timer for o/p of frequency 1 kHz and duty cycle of 75%.

Note:

design a astable using

IC-555.

Duty cycle $> 50\%$.

Duty cycle $< 50\%$.

$$T_{ON} = 0.69 (R_A + R_B) C$$

$$T_{ON} = 0.69 R_A C$$

$$T_{OFF} = 0.69 R_B C$$

$$T_{OFF} = 0.69 R_B C$$

if $D = 50\%$, then $R_A, R_B = R$.

Sol:

Given, $f = 1 \text{ kHz} = 10^3 \text{ Hz} \Rightarrow T = 1 \text{ ms}$

$$D = 75\% = \frac{75}{100} = 0.75$$

$$D = \frac{T_{ON}}{T} \Rightarrow T_{ON} = T \times D = \frac{D}{f}$$

$$= \frac{0.75}{10^3}$$

$$T_{ON} = 0.75 \times 10^{-3} \text{ s}$$

$$T = T_{ON} + T_{OFF}$$

$$T_{OFF} = 1 \text{ ms} - 0.75 \text{ ms}$$

$$T_{OFF} = 0.25 \text{ ms.}$$

$$D = \frac{(R_A + R_B)C}{(R_A + 2R_B)C} = 0.75$$

$$R_A + R_B = 0.75 R_A + 1.5 R_B$$

$$0.25 R_A = 0.5 R_B$$

$$R_A = 2 R_B$$

to find R_A & R_B :

$$T_{ON} = 0.69 (R_A + R_B)C = 0.75 \times 10^{-3} \text{ s.}$$

$$(\because R_A = 2 R_B)$$

$$(2 R_B + R_B)C = 1.0869 \times 10^{-3}$$

$$3 R_B C = 1.0869$$

$$\text{Let } C = 0.1 \times 10^{-6} \text{ F.}$$

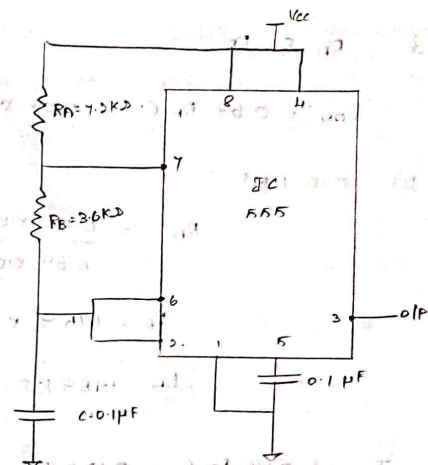
$$R_B = \frac{1.0869 \times 10^{-3}}{0.3 \times 10^{-6}}$$

$$R_B = 3.623 \times 10^3 \Omega$$

$$\therefore R_A = 2 \times R_B$$

$$R_A = 7.246 \times 10^3 \Omega$$

$$\therefore R_A = 7.24 \text{ K}\Omega, R_B = 3.623 \text{ K}\Omega$$



Design an astable multivibrator using 555 timer to produce 5 KHz o/p waveform with duty cycle of 40%.

Sol:

Given,

$$f = 5 \text{ KHz} \Rightarrow T = 0.2 \text{ ms.}$$

$$D = 40\% = \frac{40}{100} = 0.4$$

$$D = \frac{T_{ON}}{T} \Rightarrow T_{ON} = T \cdot D = 0.4 \times 0.2 \times 10^{-3}$$

$$T_{ON} = 0.08 \text{ ms.}$$

$$T = T_{ON} + T_{OFF}$$

$$T_{OFF} = (0.2 - 0.08) \times 10^{-3}$$

$$T_{OFF} = 0.12 \text{ ms.}$$

to find R_A & R_B .

$$T_{ON} = 0.69 R_A C = 0.08 \times 10^{-3}.$$

Let $C = 0.1 \mu F$.

$$R_A = \frac{0.08 \times 10^{-3}}{0.69 \times 10^{-4}}$$

$$R_A = 1.159 \times 10^3 \Omega.$$

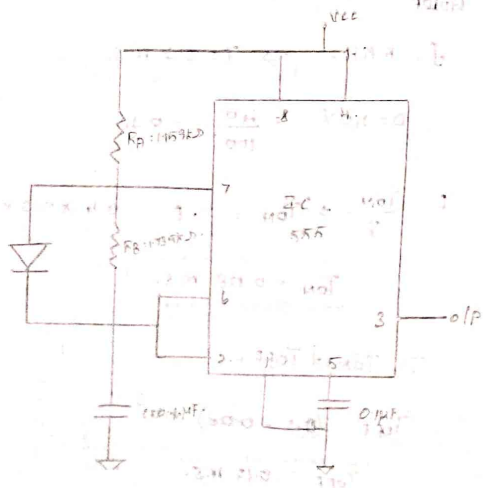
$$R_A = 1.159 K\Omega.$$

$$T_{OFF} = 0.69 R_B C = 0.12 \times 10^{-3}.$$

Let $C = 0.1 \mu F$.

$$R_B = \frac{0.12 \times 10^{-3}}{0.69 \times 10^{-4}}$$

$$R_B = 1.739 K\Omega.$$



Design an astable multivibrator using IC 555 to generate a o/p of 5 kHz. with duty cycle of 50%.

Sol:

Given,

$$f = 5 KHz. \Rightarrow T = 0.5 ms.$$

$$D = 50\% = \frac{50}{100} = 0.5$$

$$D = \frac{T_{ON}}{T} \Rightarrow T_{ON} = D \cdot T = 0.5 \times 0.5 \times 10^{-3}$$

$$T_{ON} = 0.25 ms.$$

$$T = T_{ON} + T_{OFF}$$

$$T_{OFF} = (0.5 - 0.25) \times 10^{-3}$$

$$T_{OFF} = 0.25 ms.$$

To find R_A & R_B .

$$T_{ON} = 0.69 R_A C = 0.25 \times 10^{-3}$$

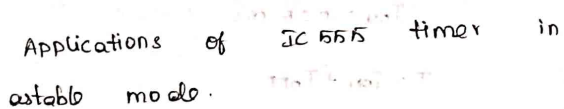
Let $C = 0.1 \mu F$.

$$R_A = \frac{0.25 \times 10^{-3}}{0.69 \times 10^{-6}}$$

$$R_A = 3.623 K\Omega.$$

Similarly,

$$R_A = R_B = 3.623 K\Omega.$$



free running ramp generator (x-)
and its derivation

$\propto \frac{1}{\omega}$
 o/p frequency $\propto$ strongly depends
 on i/p voltage

controlled by voltage applied

↓
Manufacturer
Name

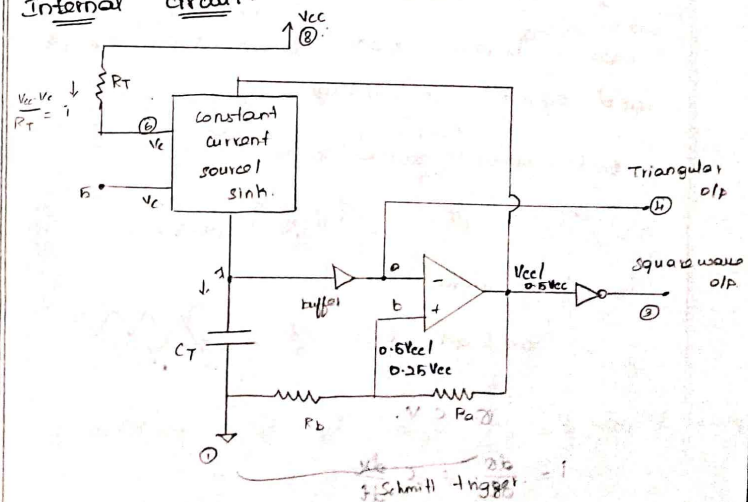
Hand-drawn circuit diagram of a 555 timer in monostable mode. The 555 timer is shown in a rectangular box with pins 1 through 8. Pin 1 is connected to GND. Pin 2 is connected to Vcc. Pin 3 is connected to Vcc. Pin 4 is connected to Vcc. Pin 5 is connected to Vcc. Pin 6 is connected to Vcc. Pin 7 is connected to Vcc. Pin 8 is connected to Vcc. The output of the timer (pin 14) is connected to a square wave pulse. The timing network consists of a resistor (R_T) and a capacitor (C_T) connected between pins 6 and 7. The control voltage (pin 5) is connected to a control voltage source.

$H \rightarrow$ " " " " triangle " "

$C_T \rightarrow$ Timing capacitor

$C_T \rightarrow$ Timing capacitor

Internal circuit:



conditions:

1. Output of Schmitt trigger vary V_{cc} to $0.5 V_{cc}$.

$$V_T = \frac{V_{cc} \cdot R_b}{R_a + R_b} = 0.5 V_{cc} \quad [R_a = R_b]$$

o/p of ST = V_{cc}

$$V_T = \frac{0.5 V_{cc} R_b}{R_a + R_b} = 0.5 V_{cc}$$

o/p of ST = $0.5 V_{cc}$

* when o/p of ST is V_{cc} , capacitor charges, when beyond $0.5 V_{cc}$ charges rise, the o/p goes $0.5 V_{cc}$.

* The capacitor discharges, when it goes below $0.5 V_{cc}$, o/p changes V_{cc} and again it charges.

without constant current source,

o/p will be



by using output source.

we get binary o/p



$$Q = C V$$

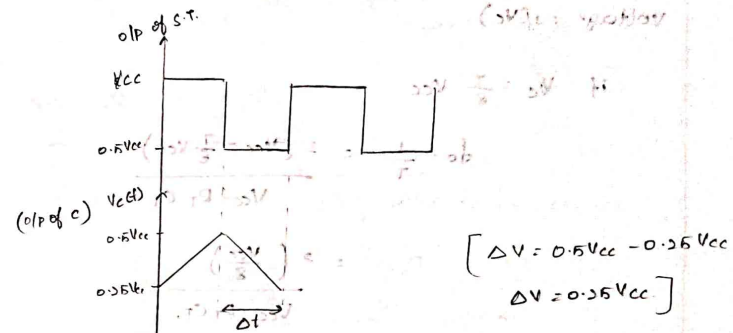
$$i = \frac{dQ}{dt} = C \frac{dV}{dt}$$

$$i = C \frac{\Delta V}{\Delta t}$$

This relates the current & voltage difference of capacitor.

$$i = \frac{V_{cc} - V_c}{R_T}$$

Relating voltage difference of capacitor.



$$\frac{V_{cc} - V_c}{R_T} = \frac{C_T [0.5 V_{cc}]}{\Delta t}$$

$$\Delta t = \frac{0.5 V_{cc} \cdot C_T \cdot R_T}{(V_{cc} - V_c)}$$

$$T = \Delta t + \Delta t = 2 \Delta t = \frac{0.5 V_{cc} C_T R_T}{(V_{cc} - V_c)}$$

$\Rightarrow$ It takes Δt for charging & Δt for discharging

$$f_0 = \frac{1}{T} = \frac{2(V_{CC} - V_C)}{V_{CC} \cdot R_T \cdot C_T} \rightarrow (1)$$

General expression of V_C .

Here, all are constant (V_{CC} , R_T , C_T) except V_C - control voltage.

The ϕ P frequency is controlled by voltage (V_C)

$$\text{if } V_C = \frac{1}{8} V_{CC}.$$

$$f_0 = \frac{1}{T} = \frac{2(V_{CC} - \frac{1}{8} V_{CC})}{V_{CC} \cdot R_T \cdot C_T}$$

$$= \frac{2 \left(\frac{V_{CC}}{8} \right)}{V_{CC} \cdot R_T \cdot C_T}$$

$$f_0 = \frac{0.25}{R_T C_T} \rightarrow (2)$$

Voltage-frequency conversion ratio (K_V):

To change the frequency we should change the V_C .

$$f_1 = \frac{2(V_{CC} - V_C + \Delta V_C)}{R_T C_T V_{CC}}$$

$$\text{W.K.T } f_0 = \frac{2(V_{CC} - V_C)}{R_T C_T V_{CC}}$$

$$f_1 - f_0 = \Delta f = \frac{2V_{CC} - 2V_C + 2\Delta V_C}{V_{CC} \cdot R_T C_T} - \frac{2V_{CC} - 2V_C}{V_{CC} R_T C_T}$$

$$\Delta f = \frac{2\Delta V_C}{V_{CC} R_T C_T}$$

$$\frac{\Delta f}{\Delta V_C} = \frac{2}{R_T C_T V_{CC}} \rightarrow (3)$$

Sub (2) in (3).

$$\frac{\Delta f}{\Delta V_C} = \frac{2}{\frac{0.25}{f_0} V_{CC}}$$

$$\therefore \frac{\Delta f}{\Delta V_C} = \frac{8f_0}{V_{CC}} \Rightarrow K_V$$

Phase-Locked Loop: (PLL) ref. pdl (16/3/20)

It can able to extract the i/f signal.

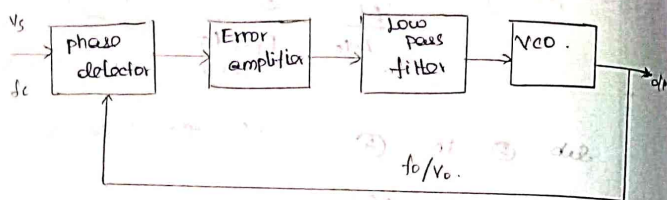
(or) It is used to fetch transmitter frequency when i/p is applied.

$\Rightarrow$ when difference b/w to $2f_s$ is 90° then the phase detector will be 0 error voltage

* Main blocks of PLL. (HF/SE 565)

- 1) Phase detector.
- 2) Error amplifier.
- 3) Low pass filter.
- 4) VCO.

Block diagram.



* PLL can be operated in 3 Modes.

1. Free Running Mode. (i/p is not given)
2. Capture Mode. - capture Range
3. Lock Mode. (capture Range not shown)

Pull-in Time

In Free Running Mode the PLL act as VCO.

In remaining 2 modes are used to find when the o/p freq of VCO is equal to transmitted freq.

$$f_o = f_c$$

error amplifier - It depends upon o/p of phase detector (together i/p signal)

phase detector - Generally a multiplier circuit

o/p is $(f_s + f_o)$ sum of 2 freq
 and $(f_s - f_o)$ diff. of two freq

Capture mode $\rightarrow$ PLL starts to track applied i/p.

Lock mode $\rightarrow$ PLL can exactly maintain the same relationship $f_o = f_s$ (i/p freq)

capture range - over a period of time when it is tracking the i/p signal.

Lock-in range $>$ capture Range, $\Delta f_1 > \Delta f_c$

For better performance

pull-in time: Time taken by the PLL to reach (attain) the lock mode

UNIT-8.

Analog to Digital and Digital to Analog Converters:-

Advantage of processing digital signal over analog signal.

①. accuracy is more.

②. less Noise

Noise Immunity:

The amount of Noise added to system without affecting its performance.

③. More noise immunity.

④. Storage is easy / Retains its originality of signal for long time.

⑤. Reconfigurable.

⑥. use of DSP algorithm is more.

⇒ In DSP it allows plenty of algorithm to process the digital data whereas in analog it has less algorithm.

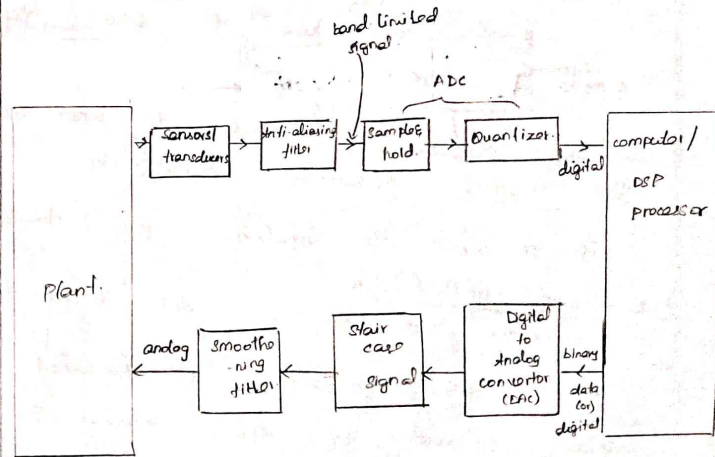
Transducers → convert Non electrical to electrical signal.

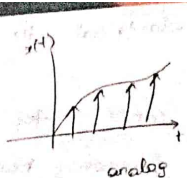
Signal has to come for only certain limit of frequency - bandlimited freq.

Anti-aliasing filter - its output should be bandlimited signal.

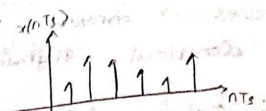
To convert Analog to digital the signal should be bandlimited then only we go to sampling.

sampling: It is process which can digitalize the analog signal.





analog



discrete - amplitude is continuous
digital - it has only 0's & 1's

Quantizer - It assigns the binary value to each & every amplitude of signal.
By smoothing the stair signal of 1/p (bit rate) signal, we get the approximate transmitted signal.

Sampling & Reconstruction

=> To sample we have 2 condition.

1. Nyquist condition / criterion - $f_s \geq 2f_m = f_c$

2. It says how to select the sample frequency.

$f_s \geq 2f_m$ [in order to reconstruct the original transmitted signal]

=> To sample the analog signal should be band-limited signal.
i.e. that we use anti-aliasing filter

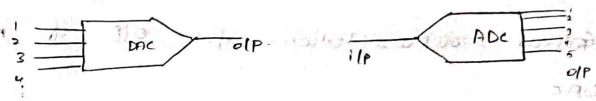
allowing avoid overlapping of adjacent spectrum because of that improper in Nyquist condition, o/p we get some aliasing

Types:

1. weighted Resistor DAC.
2. R-2R ladder type DAC.
3. Inverted R-2R ladder type DAC.

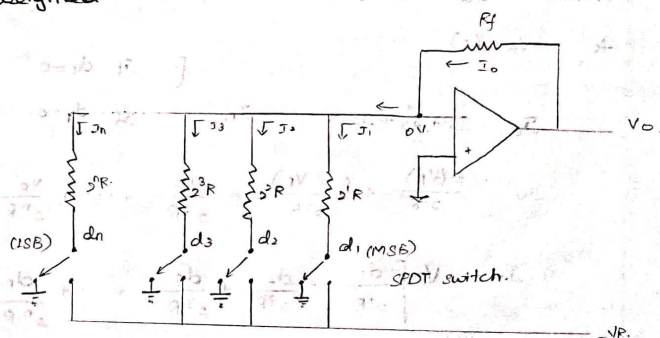
DAC:

we apply 'n' no. of binary i/p gives 1 o/p.

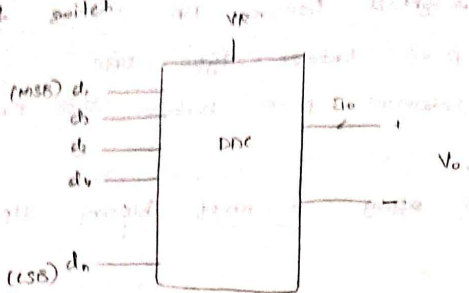


For DAC, configure op-amp as inverting summing amplifier because 'n' no. of i/p's.

1. weighted Resistor DAC.



How can we convert single bit double through switch.



General representation for O/P of n-bit DAC.

$$V_o = k V_{FS} [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n}]$$

$V_{FS} \rightarrow$ Full scale voltage.

Assume all Resistor through switch connected to $-V_R$ (r)

$$I_o = I_1 + I_2 + I_3 + \dots + I_n \quad \left[\begin{array}{l} \text{If } d_1 = 0 \quad d_1 \rightarrow \text{ground} \\ \text{If } d_1 = 1 \quad d_1 \rightarrow -V_R \end{array} \right]$$

$$I_o = \frac{0(-V_R)}{2^1 R} + \frac{0(-V_R)}{2^2 R} + \frac{V_R}{2^3 R} + \dots + \frac{V_R}{2^n R}$$

$$I_o = V_R \left[\frac{d_1}{2^1 R} + \frac{d_2}{2^2 R} + \frac{d_3}{2^3 R} + \dots + \frac{d_n}{2^n R} \right]$$

$$I_o = \frac{V_R}{R} [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n}]$$

$$\therefore V_o = I_o R_f$$

$$V_o = \frac{V_R}{R} [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n}] \cdot R_f$$

$$V_o = V_R \cdot \left[\frac{R_f}{R} \right] [d_1 2^{-1} + d_2 2^{-2} + \dots + d_n 2^{-n}]$$

Compare (1) & (2)

$$k = \frac{R_f}{R} ; V_{FS} = V_R$$

$k \rightarrow$ Gain,

If $R_f = R$ then Gain $[k=1]$

for n bit to equivalent analog.

Max value of Resistor $\rightarrow 2^n \cdot R$

Min Value of Resistor $\rightarrow 2^1 \cdot R$

Disadvantage/drawback.

* It is not suitable for large range of binary I/P's [beoz it requires large value of Resistor].

$\rightarrow$ Requires wide Range of 'R'

* well suited for only $n \leq 8$ bit.

* accuracy is less [due to Resistor].

when ON it provides nearly 0 Resistance, if OFF it provides infinite Resistance for that input should be CMOS connected.

* If BJT connected ON Resistance will not less.

The weightage of bits are

$$\text{MSB} \rightarrow d_1 = \frac{V_{FS}}{2^1}$$

$$\text{LSB} \rightarrow d_n = \frac{V_{FS}}{2^n}$$

drawback:

1. wide range of R requires.

2. very careful in selection of switch

3. Fabrication of resistance is complex.

4. accuracy is less

→ The selection of R is $2.5 \text{ K}\Omega$ to

avoid overloading effect.

→ Since the current the resistor (is constant) (or) not providing linear output.

→ since power dissipating involved in weighted type resistor is more (different) leads to non linear output.

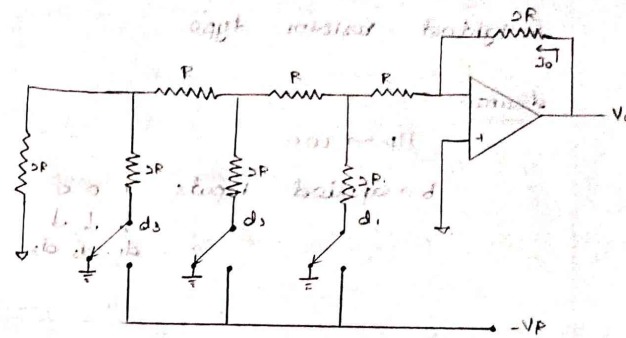
Non linearity output (due to heat produced in the circuit).

②. $R-2R$ ladder type DAC.

Here the resistors will be

either R (or) $2R$.

of ladder type is shown in figure.



$R-2R$ ladder type 3 binary I/P

assume 3-bits.

$$b = d_1 d_2 d_3$$

if $R = 2.5 \text{ K}\Omega$.

* This technique reduce the wide range of resistor.

* Fabrication is easier compare to weighted resistor type.

* Here we require only $R, 2R$ resistors.

* Current through the resistors are varied by applied inputs.

* It could not offer constant voltage

* Power dissipation & heat produced

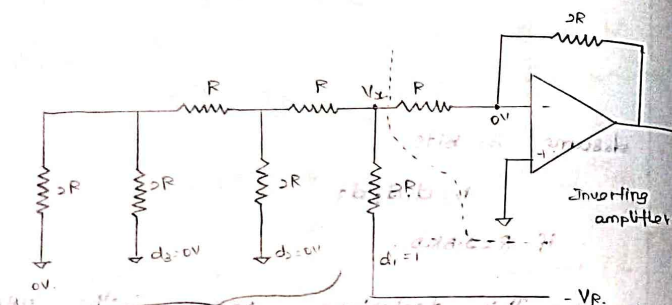
in the circuit is less compared to weighted resistor type.

assume:

$$i/p \rightarrow 100$$

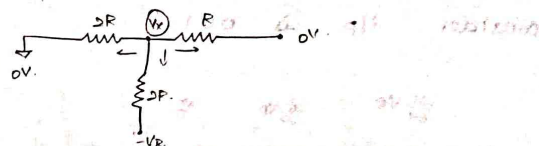
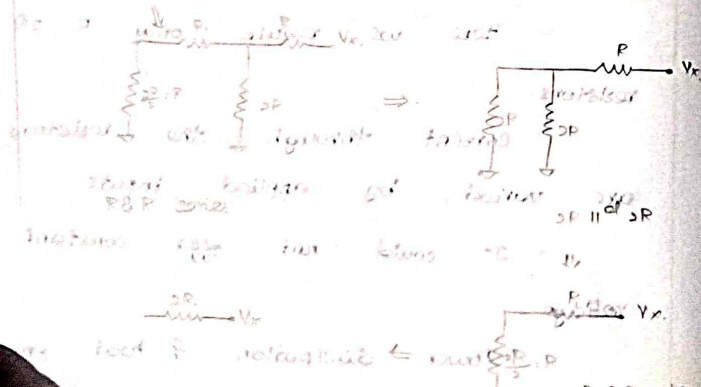
b → applied inputs

1 0 0
↓ ↓ ↓
d₁ d₂ d₃



$$V_o = -\frac{2R}{R} \cdot V_x \rightarrow \text{①}$$

2R || 2R



Apply Nodal Analysis.

$$\frac{V_x - 0}{2R} + \frac{V_x + V_R}{2R} + \frac{V_x - 0V}{R} = 0$$

$$V_x \left[\frac{1}{2R} + \frac{1}{2R} + \frac{1}{R} \right] + \frac{V_R}{2R} = 0$$

$$V_x \left[\frac{2}{2R} + \frac{1}{R} \right] + \frac{V_R}{2R} = 0$$

$$V_x \left[\frac{2}{R} \right] + V_R \left[\frac{1}{2R} \right] = 0$$

$$V_x = -\frac{V_R}{2R} \left[\frac{R}{2} \right]$$

$$V_x = -\frac{V_R}{4} \rightarrow \text{②}$$

Sub ② in ①.

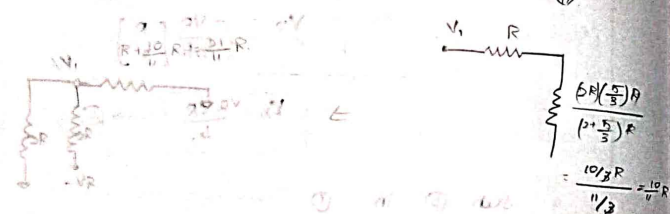
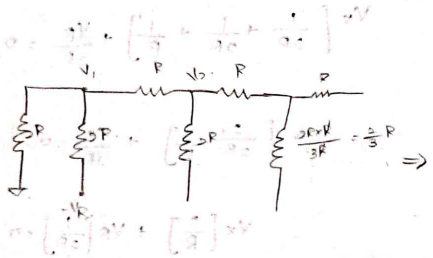
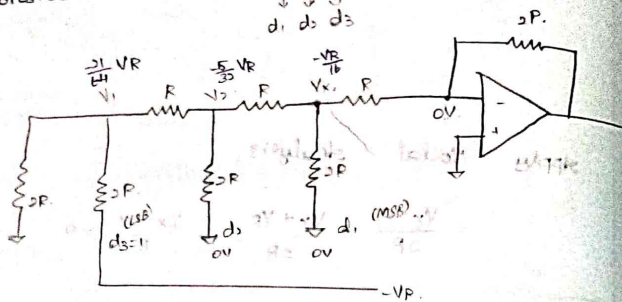
$$V_o = -2 \cdot \left(-\frac{V_R}{4} \right)$$

$$V_o = \frac{V_R}{2}$$

If $V_R = 5V$ then $V_o = 2.5V$.

consider i/p is 001.

d_1, d_2, d_3



apply Nodal analysis.

$$\frac{V_1}{2R} + \frac{V_1 + V_R}{2R} + \frac{V_1}{\left(\frac{21}{11}\right)R} = 0.$$

$$V_1 \left[\frac{1}{2R} + \frac{1}{2R} + \frac{11}{21R} \right] = -\frac{V_R}{2R}.$$

$$V_1 \left[\frac{1}{R} + \frac{11}{21R} \right] = -\frac{V_R}{2R}.$$

$$V_1 \left[1 + \frac{11}{21} \right] = -\frac{V_R}{2}.$$

$$V_1 = -\frac{V_R}{2} \left[\frac{21}{32} \right]$$

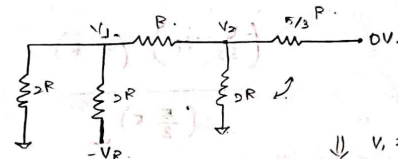
$$V_1 = -\frac{V_R (21)}{64}$$

$$V_1 = -\frac{21}{64} V_R$$

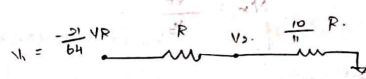
W.K.T.

$$V_0 = -\frac{2R}{R} V_x.$$

$$V_0 = -2 V_x. \rightarrow (1)$$



$$V_2 = -\frac{21}{64} V_P.$$

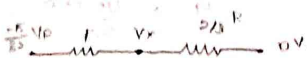
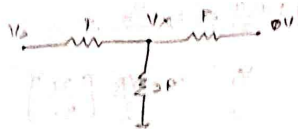


Voltage divider rule,

$$V_2 = \frac{\left(\frac{21}{64} V_R\right)}{\left(R + \frac{10}{11} R\right)} \left(\frac{10}{11} R\right)$$

$$V_2 = \frac{-\frac{21}{64} V_R \left(\frac{10}{11}\right)}{\left(\frac{21}{11}\right)}$$

$$V_2 = -\frac{10}{64} V_R. = -\frac{5}{32} V_R.$$



Voltage divider rule.

$$V_x = \left(\frac{R_2}{R_1 + R_2} V_p \right) \left(\frac{R_2}{R} \right)$$

$$= \left(\frac{R_2}{R_1 + R_2} V_p \right) \left(\frac{R_2}{R} \right)$$

$$= \left(\frac{R_2}{R} \right) V_p$$

$$V_x = -\frac{V_p}{16}$$

Sub V_x in equ (1).

$$V_o = \left(\frac{-2R}{R} \right) V_x$$

$$V_o = -2 \left(\frac{-V_p}{16} \right)$$

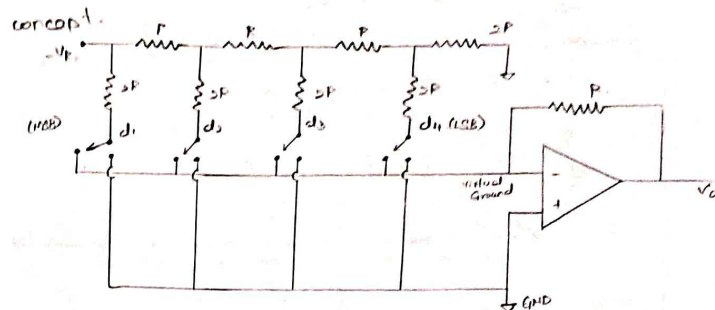
$$V_o = \frac{V_p}{8}$$

if $V_p = 5V$ then $V_o = 0.625V$.

for $b = 001$

③ Inverted Type R-2R Ladder DAC:

In this case the resistors are connected directly / indirectly to ground for both 0 and 1 using virtual ground concept.



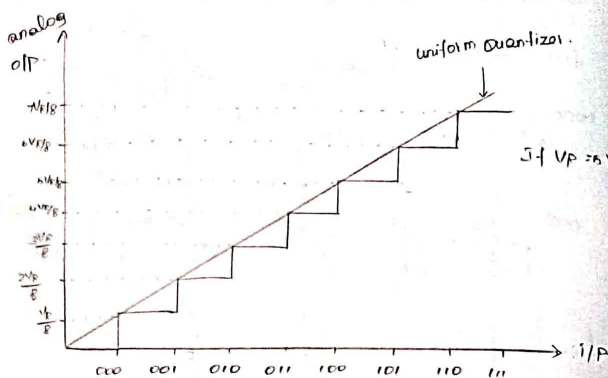
④ Difference b/w R-2R & Inverted R-2R

- * With respect to each node it provides equal resistance through the ladder of
- * The current flowing through each node is constant irrespective of applied i/p
- * LSB and MSB are exchanged.

It is also called current mode inverted R-2R ladder type DAC.

The fn of circuit is explained by summing the current in each and every node (Resistance).

→ o/p of DAC in the form of stair case signal.



Resolution (or)

Resolution (or)

1 LSB (or)

step signal size

$$= \frac{V_R}{2^N} \quad (\text{use follow this})$$

$$R = \frac{V_R}{2^{N-1}} \quad \left\{ \begin{array}{l} \text{well suited when } N \text{ is large} \end{array} \right.$$

$N \rightarrow$ bit size = 8

$$R = \text{step size} = \frac{V_R}{2^8} = \frac{V_R}{8}$$

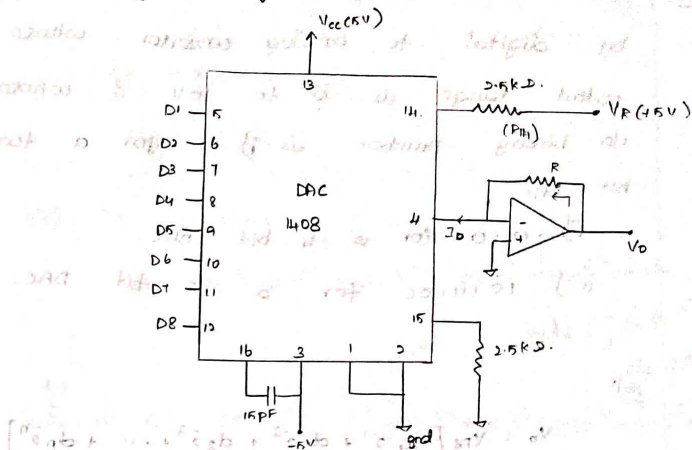
For N bit DAC the no. of step

size is $2^N - 1$

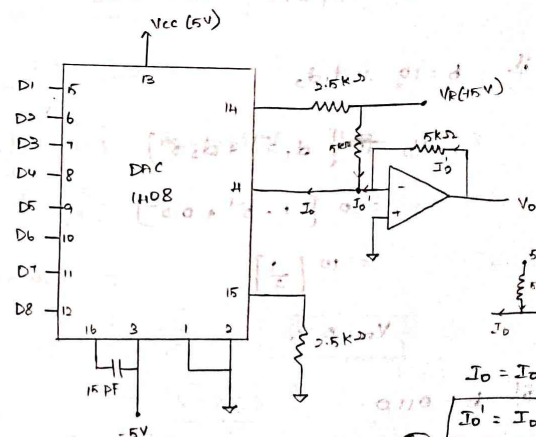
The overall output is $V_O = (V_{FS}) - 1 \text{ LSB}$

always.

Pin - Diagram of DAC - 1408 :



$$I_O = \frac{V_R}{R_{14}} \left[\sum_{i=1}^N d_i 2^{-i} \right] \text{ is obtained from pin 14}$$



$$V_O = I_O R_f \quad \text{--- (1)}$$

$$V_O = \left[I_O - \frac{V_R}{5K\Omega} \right] R_f$$

$$I_O = I_O' + 1 \text{ mA}$$

$$I_O = I_O' - 1 \text{ mA}$$

$$V_O = I_O' R_f$$

① What output voltage would be produced by digital to analog converter whose output range is 0 to 10V & whose i/p binary number is i) 10 for a two bit DAC.

ii) 0110 for a 4 bit DAC.

iii) 10111100 for a 8 bit DAC.

In IIC $V_{FS} = V_R$.

Sol:

$$V_o = V_{FS} [d_1 \overset{MSB}{2^{-1}} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n \overset{LSB}{2^{-n}}]$$

$$(or) V_o = V_R [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + \dots + d_n 2^{-n}]$$

i) $b = 10 = d_1 d_2$.

$$V_o = 10 [d_1 2^{-1} + d_2 2^{-2}]$$

$$= 10 [1 \cdot 2^{-1} + 0 \cdot 2^{-2}]$$

$$= 10 \left[\frac{1}{2} \right]$$

$$V_o = 5V.$$

ii) $b = 0110$.

$$V_o = 10 [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + d_4 2^{-4}]$$

$$= 10 [(0) 2^{-1} + (1) 2^{-2} + (1) 2^{-3} + (0) 2^{-4}]$$

$$= 10 \left[\frac{1}{4} + \frac{1}{8} \right]$$

$$= 10 \left[\frac{3}{8} \right]$$

$$V_o = \frac{15}{4} V$$

$$V_o = 3.75 V.$$

iii) $b = 10111100$.

$$V_o = V_R [d_1 2^{-1} + d_2 2^{-2} + d_3 2^{-3} + d_4 2^{-4} + d_5 2^{-5} + d_6 2^{-6} + d_7 2^{-7} + d_8 2^{-8}]$$

$$V_o = 10 [(1) 2^{-1} + (0) 2^{-2} + (1) 2^{-3} + (1) 2^{-4} + (1) 2^{-5} + (1) 2^{-6} + (0) 2^{-7} + (0) 2^{-8}]$$

$$= 10 \left[\frac{1}{2} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} \right]$$

$$= 10 [0.734375]$$

$$V_o = 7.34375 V$$

iv) Find its Resolution, LSB, MSB.

$$R = \frac{V_R}{2^N - 1}$$

For binary, i/p = 2.

$$R = \frac{10}{2^2 - 1} = \frac{10}{3}$$

$$R = 3.33$$

For 4 bits.

$$R = \frac{10}{2^4 - 1} = \frac{10}{15} = 0.666$$

For 8 bits.

$$R = \frac{10}{2^8 - 1} = \frac{10}{255} = 0.0392$$

$$1 \text{ LSB} = 2^{-n} V_{FS}$$

$$1 \text{ MSB} = 2^{-1} V_{FS}$$

For 3 bits,

$$1 \text{ LSB} = 2^{-3} \times 10$$

$$= \frac{10}{8}$$

$$= 1.25 \text{ V}$$

$$1 \text{ MSB} = 2^{-1} \times 10$$

$$= \frac{10}{2}$$

$$= 5 \text{ V}$$

For 4 bits,

$$1 \text{ LSB} = 2^{-4} \times 10$$

$$= \frac{10}{16}$$

$$= 0.625$$

For 8 bits.

$$1 \text{ LSB} = 2^{-8} \times 10$$

$$= 0.039$$

$$1 \text{ MSB} = 2^{-1} \times 10$$

$$= 5$$

For n bit of weighted resistor type DAC requires how many resistor.

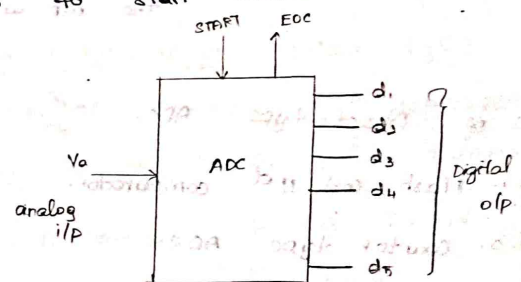
n resistor required without including R_f .

Analog to Digital converter (ADC).

* It uses two control lines

used to identify when the ADC performs the start and end of conversion

* START → which tells the ADC when to start conversion process



* EOC → End of conversion tells when the conversion process of ADC completed.

Types of ADC.

1. Direct-type ADC

* applied analog signal is compared with internally generated equivalent (equivalent internal) signal to produce the binary o/p.

2. Indirect-type ADC

* It produces the equivalent linear o/p w.r. to time & frequency then produce binary o/p.
* It modifies the i/p to equivalent linear o/p and then the o/p is produced.

Types of Direct-type ADC. (Refer slide 2006)

1. Flash (or) parallel comparator.

2. Counter type ADC.

3. Tracking / servo converter.

4. Successive approximation type ADC.

1. Flash type (or) parallel comparator.

* It is the simplest ADC.

* It is also a fastest ADC.

→ Conversion provided by this is much speed than others).

* Speed is more so conversion time is less. (100 ns / 100 ns)

→ becoz the conversion mechanism works simultaneously.

* Simultaneous conversion operation.

It has two circuits.

i) comparators.

ii) Voltage divider circuit.

For n-bit binary (Flash type).

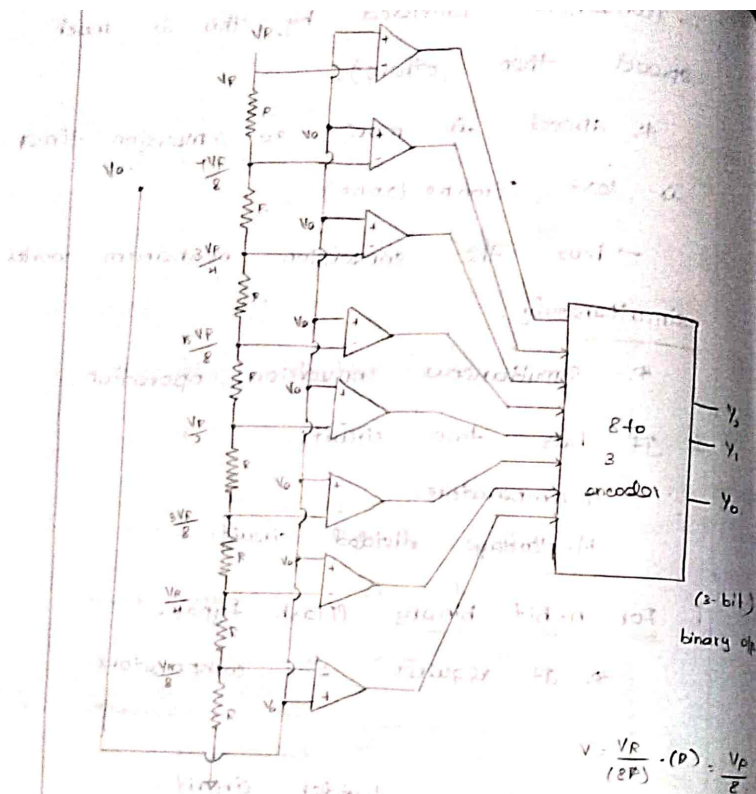
* It requires $2^n - 1$ comparators.

eg. for 3 bits $2^3 - 1 = 7$ comparators used.

* Voltage divider circuit.

→ It requires 2^n resistors.

In comparator we compare analog i/p signal & internal equivalent signal (fixed terminal).



$$V = \frac{V_R}{(2^n)} \cdot (D) = \frac{V_R}{2}$$

If $V_a > V_b$, $x=1$

$V_a < V_b$, $x=0$

$V_a = V_b$, previous state.



applied analog i/p	x_7	x_6	x_5	x_4	x_3	x_2	x_1	x_0	y_2	y_1	y_0
0 to $V_R/8$	0	0	0	0	0	0	0	1	0	0	0
$\frac{V_R}{8}$ to $\frac{V_R}{4}$	0	0	0	0	0	0	1	1	0	0	1
$\frac{V_R}{4}$ to $\frac{3V_R}{8}$	0	0	0	0	0	1	1	1	0	1	0
$\frac{3V_R}{8}$ to $\frac{V_R}{2}$	0	0	0	0	1	1	1	1	0	1	1
$\frac{V_R}{2}$ to $\frac{5V_R}{8}$	0	0	0	1	1	1	1	1	1	0	0
$\frac{5V_R}{8}$ to $\frac{3V_R}{4}$	0	0	1	1	1	1	1	1	1	0	1
$\frac{3V_R}{4}$ to $\frac{7V_R}{8}$	0	1	1	1	1	1	1	1	1	1	0
$\frac{7V_R}{8}$ to V_R	1	1	1	1	1	1	1	1	1	1	1

* It requires $2^n - 1$ comparators only since we using 2:3 encoder so we use no additional comparator (x).

* If question ask how many comparators used in flash-type structure: $2^n - 1$.

Draw back:

→ This technique is not suited for large no. of bits.

→ The requirement of comparator is doubled when one bit is increased.

2). Counter - type ADC.

* It has three blocks.

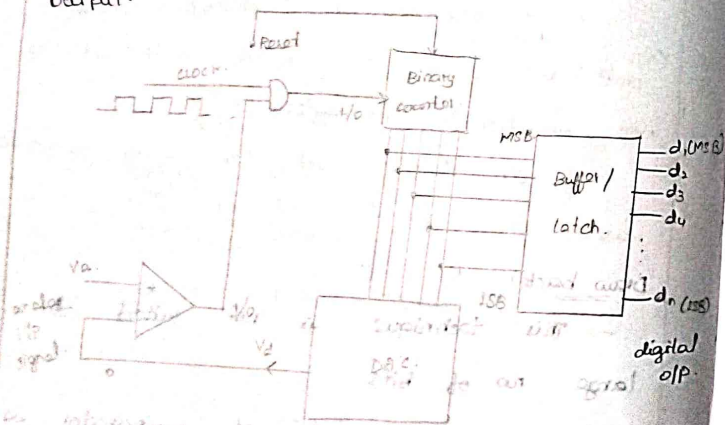
i). Comparator
→ I/P's * I/P analog signal.
* output of DAC

ii). DAC.

iii). Binary counter. (B.C)

* o/p of Binary counter is connected to DAC.

* Equivalent digital output is produced from Binary counter for the applied analog input and then using Buffer we get output.



B.C starts to count in two cases.

1. we need to release the reset (set 0)



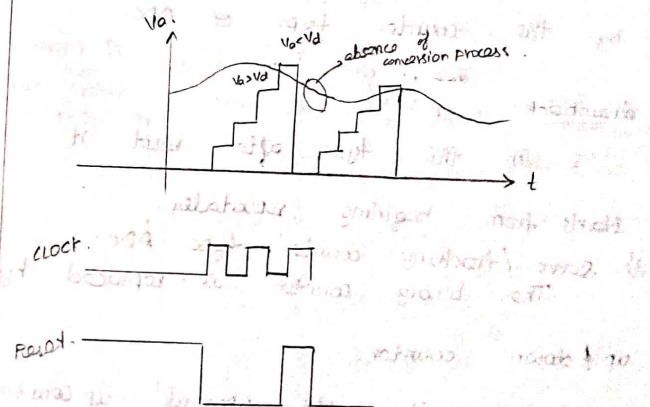
2. apply analog I/P signal

if $V_a > V_d$, o/p of comparator $\rightarrow 1$.

give clock signal to the AND gate then the B.C perform counting process. The clock signal can be passed till the $V_a > V_d$

if $V_a < V_d$, o/p of comparator $\rightarrow 0$.

Here clock signal is not passed, B.C will not count, so it is reset. after it performs repeatedly.



* Conversion time is strongly depend on the amplitude of i/p analog signal.

drawback:

1. Conversion time is more.

2. Speed is less.

→ o/p of ADC depends on.

* when counter Reset. (Resetting Processing of counter)

* o/p of DAC.

Conversion time:

$$T_c = (2^N - 1) T_{clk}$$

It is the conversion time offered by the counter type of ADC.

drawback: There is chance to absence of conversion process the after reset & before start of other clock

3. In this type, after reset it

starts from beginning repeatedly.

3). Servo / Tracking counter type ADC:

The binary counter is replaced by up & down counter.

when it counts upward - up counter.

($V_a > V_d$)

when it counts downward - down counter.

By changing the behaviour of counter we can reduce the conversion time.

Less conversion time compared to Counter type.

1). Successive approximation Type ADC.

If $V_a > V_d$ → the MSB bit is to be retained and the bit in MSB is set (1) (o/p of c → 1)

If $V_a < V_d$ → make consent bit as 0 make next bit as 1. (o/p of comparator → 0)

11010100 → is the digital equivalent of applied analog i/p signal.

1st give start signal.

V_a	o/p of comparator	V_d (o/p of SAR)	
11010100	1	10000000	$V_a > V_d$
	1	11000000	[MSB same]
	0	11100000	$V_a > V_d$ next bit → 1
	1	11010000	$V_a < V_d$ MSB → 0 next → 1
	0	11011000	$V_a > V_d$
	1	11010100	$V_d > V_a$

eg:

Resolution: 4-bit.

$$V_{ref} = 8V.$$

$$V_{in} = 5.2.$$

for 1000,

$$\text{DAC output, } V_o = V_R [d_1 2^{-1} + \dots + d_n 2^{-n}]$$

$$= 8 \times \frac{1}{2}$$

$$V_o = 4V.$$

$$5.2V > 4V.$$

keep.

Next:

for 1100

$$\text{DAC O/P } \Rightarrow V_o = 8 \left[\frac{1}{2} + \frac{1}{4} \right] = 6V.$$

$$V_a < V_d$$

$$5.2 < 6.$$

Reset.

then, 1010

$$\text{O/P DAC, } V_d = 8 \left[\frac{1}{2} + \frac{1}{8} \right] = 5V.$$

$$5.2 > 5V.$$

keep MSB.

then 1011

$$V_d = 8 \times \left[\frac{1}{2} + \frac{1}{8} + \frac{1}{16} \right] = 5.5V.$$

$$5.2 < 5.5 \quad \text{Reset.}$$

then 1010

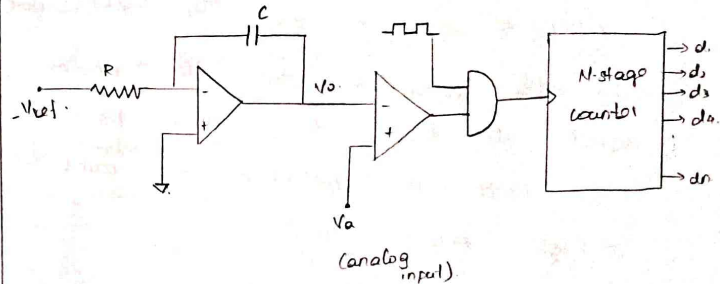
The logic has to be tested for all bits of 8AB with the equivalent digital representation.

Indirect type ADC / Integrating type ADC.

types:

1. charge balanced ADC / single slope ADC.
2. Dual slope ADC.

①. Charge - balanced ADC:

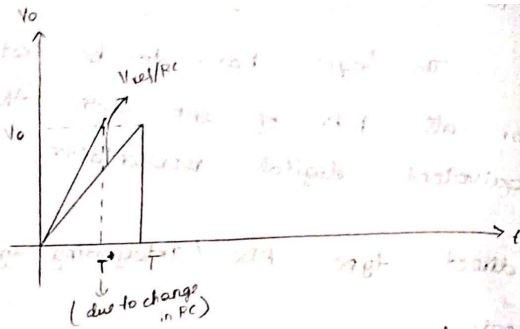


$$|V_{ref}| > V_a.$$

The magnitude of V_{ref} should be greater than V_a .

$$V_o = \frac{1}{Rc} \int -V_{ref} \cdot dt.$$

$$V_o = \frac{V_{ref}}{Rc} \cdot T$$



The output of integrator is connected to comparator circuit.

When $V_o < V_a$, o/p of comparator $\rightarrow 1$ (V_a)

(o/p of comp is given to AND gate)

By this converting the i/p analog signal to linear time variant signal which is connected to counter

we get binary o/p

When $V_o > V_a$, o/p of comparator $\rightarrow 0$ (V_a)

at $t = t_2 = 0$

Since $V_a > V_o$

The integrator integrates the

V_{ref} until the o/p of comparator

is $V_o < V_a$

The moment when $V_o > V_a$, o/p of $C \rightarrow 0$ counter stops the counting process.

* If RC is less for the same analog i/p I get time reduced.

draw back:

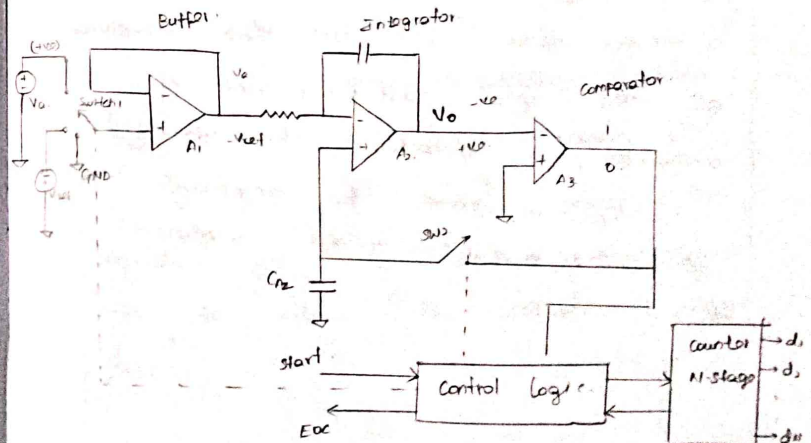
* The same analog i/p by change in RC we get linear time variant signal is varied.

* Linearly ~~changing~~ ^{time} changing signal

depends on RC for applied analog signal.

* Poor accuracy. $\downarrow$ less %

⑤. Dual-slope ADC



There are three types of op-amp

circuit is used,

1. Buffer (A1)

2. Integrator (A2)

3. Comparator (A3)

⇒ 1st stage - switch 1 is closed, it

allows non-inverting terminal to ground

SW 2 is closed

It occurs before the starting

process of ADC.

[means we don't apply the i/p to integrator]

⇒ without applying i/p, there is presence of i/p offset voltage

→ for that we use C_{A2} . it stores all the offset volt (bec of op of comparator is connected to capacitor)

C_{A2} is used to compensate the effect of i/p offset voltage appeared in all 3 circuits

before
START

①. SW1 → END, SW2 → closed.

CA → compensate the i/p offset volt present in A1, A2, A3

②. SW2 → open, START.

SW1 → 'Vo'

o/p of integrator, $V_o = -\frac{1}{RC} \int V_a dt$

 $T_c = \text{clock time period}$

Since V_o is negatively varying slope signal.

so o/p of comparator is always → 1.

which allows the counter to

count upto 0 to $2^N - 1$

[during the integration process the counter will count upto $2^N - 1$].

At end of counting process. the counter gives the signal to control logic to allow the SW1 to connect $-V_{ref}$.

$$V_a = -\frac{V_o}{RC} T_i; T_i = 2^N T_c$$

Stage-3

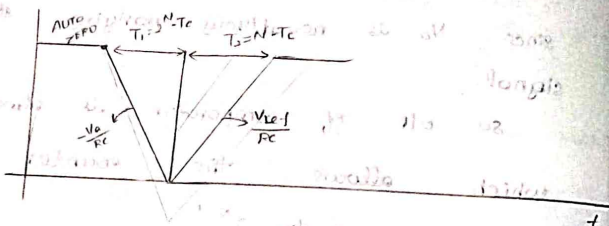
(3) SW1 $\rightarrow -V_{ref}$.

$$\begin{aligned} \text{o/p of integrator} &= \frac{1}{RC} \int -V_{ref} dt. \\ &= \frac{V_{ref}}{RC} T_3. \end{aligned}$$

we get $-V_0$ Ramp.

$\therefore$ The o/p of comparator is 0.
[The moment at crossing 0]

$$V_- > V_+$$



when o/p of C is 0 the counter stops counting that time period is

$$T_3 = N \cdot T_c$$

we add the two signal,

linearly changing time w.r. to analog signal.
& linearly V_{ref} .

$$\frac{-V_0}{RC} T_1 + \frac{V_{ref}}{RC} T_3 = 0.$$

$$\frac{V_{ref}}{RC} T_3 = \frac{V_0}{RC} T_1$$

$$T_3 = \frac{V_0}{V_{ref}} \cdot T_1$$

$\rightarrow$ Time period is depends on V_0, V_{ref} not the RC.

$\rightarrow$ if we change V_0 T_3 also changes
 $V_0 \uparrow \Rightarrow T_3 \uparrow$.

sub the value of T_1 & T_3 .

$$N \cdot T_c = \frac{V_0}{V_{ref}} \cdot N \cdot T_c$$

(X)

$$V_0 = \frac{N V_{ref}}{2^N}$$

A dual slope ADC uses a 16-bit counter and a 4 MHz clock rate the max. i/p voltage is +10 Volt, the max integrator o/p voltage should be -8V when the counter has cycled through 2^N counts. The capacitor used in integrator is 0.1 μF . Find the

Value of R in the integrator.

Sol:

$$V_a = +10V.$$

$$N = 16 \text{ bit.}$$

$$(V_i) V_o = -8V.$$

$$C = 0.1 \mu F.$$

$$\text{Clock rate } f = 4 \text{ MHz.}$$

$$T_c = \frac{1}{4 \times 10^6}.$$

$$T_c = 0.25 \times 10^{-6}.$$

$$T_c = 0.25 \mu\text{sec.}$$

$$-8V = \frac{-V_a}{R} T_c.$$

$$-8 = \frac{-V_a}{R} \times 2^N \times T_c.$$

$$8 = \frac{10}{R \times 0.1 \times 10^{-6}} \times 2^{16} \times 0.25 \times 10^{-6}.$$

$$R = \frac{10}{8 \times 0.1} \times 65,536 \times 0.25$$

$$= \frac{10}{0.8} \times 16384.$$

$$= \frac{163840}{0.8}$$

$$R = 204800.$$

$$R = 204.8 \text{ k}\Omega.$$

$$R \approx 205 \text{ k}\Omega.$$

$$V_a = \frac{N V_{ref}}{2^N} \quad (*)$$

$$V_{ref} = \frac{10 \times 2^{16}}{16}.$$

$$= \frac{10 \times 65536}{16}.$$

$$= \frac{655360}{16}.$$

$$V_{ref} = 40960 \text{ V}$$

if 3-bit,

$$V_a = \frac{10 \times 2^3}{8}.$$

$$V_a = \frac{80}{8} = 10 \text{ V.}$$

Advantage:

→ It could provide best accuracy (high)

drawback:

* It is only good for slowly varying signal.

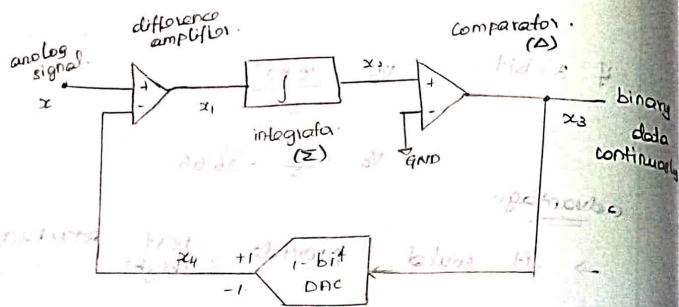
* Conversion time is more.

Delta-sigma type ADC: 10/01 PPT 1/11/2026

Δ → represents the difference (or) change in parameter in generally.

Δ → represents the comparator.

Σ → $\int$ of sigma is replaced as an integrator circuit.



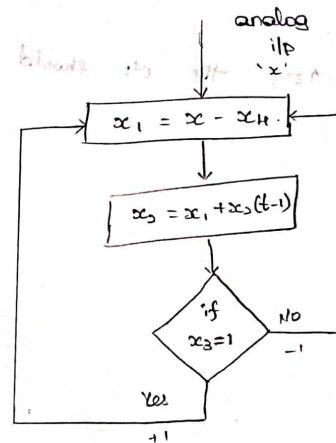
→ the o/p of DAC is either +1 (or) -1 designed.

* If it receives 1 as input → o/p of DAC +1

* If it receives 0 as input → o/p of DAC -1

Integrator - simply performs addition operation.

The entire function can also be represented in flowchart:



x → applied analog sig.

$x_2(t-1)$ → previous output.

$x_{H.}$ → o/p of DAC.

x_3 → o/p of integrator.

If $x_1 \rightarrow +ve$, o/p → group of 1's.

If $x_1 \rightarrow -ve$, o/p → group of 0's.

ADC IDAC specifications:

1. Resolution
2. Linearity.
3. Accuracy.
4. Settling time.
5. Stability.

① Resolution.

It tells how the output changes when 1 LSB change in i/p.